

Analysing the sustainability of community-managed water systems in Nepal

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Abstract: A multi-faceted, multi-factorial interaction of technical, financial, social, and institutional factors limits the sustainability of community-managed water systems (CMWS) in Nepal over the long run. This paper goes beyond the descriptive analysis to utilise a structural equation modelling (SEM) framework as an empirical test to quantify the association between these key determinants. The sample size used was 360 households and 36 water committee members in three different ecological regions of Nepal which were chosen through a stratified random survey. It defined latent constructs of financial viability, institutional governance, technical capacity, and community social capital, and it tested their direct and indirect impact on the measured outcome of system sustainability (measured through functionality, service quality, and user satisfaction). Findings of the SEM analysis indicate that the most powerful direct predictors of sustainability are financial viability ($b = 0.38$, $p < 0.01$) and institutional governance ($b = 0.42$, $p < 0.01$). Moreover, community social capital has a great indirect influence as it reinforces governance and financial management. The model offers a strong evidence-based model that explains more than 65% of the sustainability outcomes. The proposed study finds that the interventions should be coordinated; the key to making the rural water supply in Nepal sustainable is to invest in governance capacity and the financial system, and to foster social cohesion.

Keywords: community participation, financial viability, governance, Nepal, structural equation modelling (SEM), sustainability, water management

INTRODUCTION

Clean and safe water is not only a basic human right but also an essential factor influencing population health and socio-economic growth as well as the general well-being. Community-managed water systems (CMWS) are now the new model of providing this vital service to the rural and peri-urban residents in many regions of the Global South, especially in mountainous countries such as Nepal (Clement, Suhardiman and Bharati, 2017). These systems are a major departure from state-based provision to a decentralised provision that relies on local ownership, participatory governance, and community financing. After the failure or inefficiency of public and private water-supply systems, such

community-managed water systems have been proven as the ways of decentralisation and good governance (Adams and Zulu, 2015). Nonetheless, the sustainability of these systems on a long-term basis is a remarkably challenging task that keeps on demanding sustained investment and work on the part of the government and non-governmental organisations (Ching, 2018).

The technical, financial, social, and institutional aspects of the context within the country are intertwined, as the sustainability crisis of CMWS in Nepal is based on a complicated combination of these elements. The rugged terrain of Nepal and its dispersed communities render centralised water supply systems neither economically nor technically feasible, and CMWS serves as the key to rural water security. Yet, despite their

importance, these systems are becoming more vulnerable. Institutional aspects like elite capture, absence of transparency in water user committees (WUCs), and ineffective accountability systems undermine the trust of the community. Economically, poor tariff systems, lack of consistency in the collection of payments, and lack of funds to carry out major repairs are a menace in most systems. The socially high outmigration rates, especially of men, deprive communities of workers and leadership, but also modify the gender relations and, consequently, create pressure on the traditional social capital (Bhawana, 2020). Climate change also creates new vulnerabilities, including unpredictable precipitation, melting glaciers, and the rise in the frequency of droughts and floods, affecting the balance of water sources and infrastructure built based on the historical situation (Fatima *et al.*, 2025).

This is a very complex issue that is not particular to Nepal but occurs in its own context. In Nepal, water is still treated as a free good in many places. However, there seems to be a scarcity of water in some urban areas, like in Kathmandu, Palpa and some high hills. Investment in the water supply mechanism and completion of such an investment project are taking much time due to a lack of sufficient finance. Universal difficulty in getting financial viability is also emphasised in studies of Pakistan and Ethiopia, as it is a delicate balance between affordable tariffs and enough revenue to cover operations and maintenance (Ahmed Memon, 2004). On the same note, studies of common-pool resources around the world, including that of the Nepal community forestry sector, have continuously reiterated that effective, inclusive, and transparent governance is the foundation of successful collective action (Mwangi, 2017). Although these separate observations are important, a critical gap can be seen in the interaction between these various factors – financial, institutional, social, and technical and how they all work together to affect sustainability in the specific socio-ecological situation in the CMWS of Nepal.

The proposed work will fill this gap by going beyond descriptive and siloed analysis by empirically modelling the interrelationships between these determinants of importance. To do so, we are using structural equation modelling (SEM), an influential multivariate method that was chosen due to its capability to jointly test complex and intertwined relationships between a series of latent constructs (e.g., financial viability, social capital) and their direct and indirect interdependency with one central outcome (system sustainability). Our main hypothesis is best suited to be tested through this approach, which will allow us to determine the truth that sustainability is not caused by single factors but by a complex of institutional, financial, and social interactions.

To be representative of our findings, the study was carried out in the four locations, well selected in the Middle Hills region of Nepal. The sampling was stratified in terms of these two variables, which are: 1) ecological zone (mid-hill vs river valley) to capture the different environmental pressures, 2) migration intensity (high vs low) to learn about the effects of this significant demographic change. We have this stratified design because we want a sample that is representative of the different settings in which CMWS exists; hence, this will increase the generalisability of our results.

The sustainability of community-managed water systems (CMWS) is not only a technical issue but a socio-ecological

riddle. Centralised water provision can be seen as impractical in rugged topography and scattered settlements (such as in Nepal), so CMWS form the mainstay of rural and peri-urban water security. But their functionality is unsafe in the long term. The review encompasses an overview of the literature in order to develop a multi-dimensional framework that can explain the determinants of CMWS sustainability. It goes beyond engineering paradigms to the complex interaction of institutional governance, financial feasibility, technical and environmental considerations, socio-cultural forces and external influences that jointly make or break a system.

Integrated water resources management (IWRM) has played a huge role in the discourse about water management in Nepal. But, as Clement, Suhardiman, and Bharati (2017) suggest, IWRM has at times remained an institutional holy grail in the Nepalese context, a conceptually compelling but practically ineffective model which has failed to provide equitable and effective water justice. Such a discrepancy between a policy and local reality evidences the urgency of local governance structures.

Sustainable common-pool resource management is all about effective governance. In the case of CMWS, this can be translated to the performance of the water user committee (WUC), the main institution that handles daily activities, financial affairs, and conflict resolution. Transparency, accountability, and inclusive participation are non-negotiable principles of good governance and have constantly been pointed out in the literature (Mwangi, 2017). It is more likely that the capacity of the WUC to govern and enforce rules, provide fair distribution, and keep the community trusted is a better indicator of sustainability than the actual physical infrastructure. This is supported by the research on community forestry in Nepal, a similar common-pool resource regime, which indicates that effective collective action is directly connected to strong, locally legitimised institutions.

One of the major governance issues is the elite capture phenomenon, where the majority of the influential members of the community dominate decision-making to serve their interests at the disadvantage of other marginalised groups (women, lower castes, and poor). This compromises the perceived justice of the system, as well as undercuts the sense of community ownership that is needed to maintain the system and contribute to its funding (Roy and Shah, 2002). In addition, Ching (2018), in Kathmandu, finds a paradox of social resilience in that strong social tie, far from helping cooperation, can occasionally be understood through delays in the provision of infrastructure as communities negotiate over the complex social terms and power dynamics. This implies that social capital is a two-edged sword that can empower or cripple effective governance.

A water system can become well-controlled and technically correct but lack a safe financial base; it will be unsuccessful. Financial viability is the capability of a system to produce revenue that is adequate to sustain its operations and maintenance (O&M) expenses and to save money to have capital to conduct large-scale repairs and replacements.

The article by Ahmed Memon (2004) on the water and sanitation sector in Pakistan offers an invaluable framework that highlights the fact that economic feasibility is a condition for the sustainability of environmental services. This is by setting the right tariff frameworks that can strike a balance between affordability among the poorest consumers and the necessity to make sufficient revenues. A study by Fentahun and Gashaw

(2014) on the demand of households in Ethiopia to receive a better water supply confirms that willingness to pay is positively related to perceived quality and reliability of the service received, but is limited by household income. This presents a fine walking line to WUCs: by keeping the tariffs at a level that is too low, it becomes financially unsustainable, whereas raising them is the only way to avoid non-payment by vulnerable households.

Other than collecting tariffs, it is the wise utilisation of such funds that matters most. The most effective systems are those that do not view water as a free resource but as an economic good with specific guidelines on how funds should be managed, audited, and used to repair the system in the future (Hangan *et al.*, 2022). The absence of a sinking fund to replace a major asset is a pitfall that causes the collapse of a system when one of the major components of the system, like a pump or a reservoir tank, breaks down. This fiscal planning demands some administrative ability, which might be lacking in volunteer-based committees, and thus the necessity of continued external assistance and training.

The sustainability of any infrastructure is naturally connected to the technical design and the environment where it works. In Nepal, extreme geo-climatic diversity in the country has a very strong impact on this. The systems should be made resistant to landslides, earthquakes, and seasonal changes in the yield of the source of water.

The literature on climate change adaptation is very much applicable in this case. Agricultural settings in Bangladesh (Alam, 2015) and studies in Africa (Below *et al.*, 2012) show that households and communities shift their resource use patterns in order to respond to the environmental stressors. In the case of CMWS, climate change triggers increased vulnerability because of changes in the patterns of precipitation, melting of the glaciers, and increased droughts and floods (Fatima *et al.*, 2024). A system that is developed based on historical information on water flow can be rendered useless, compelling communities to move further / go back to unsafe sources; thus, the social contract of CMWS is compromised.

Technical sustainability is also based on the issue of local skills and access to spare parts. Mechanised systems designed to require specialist technicians and imported components often do not pass the test of survivability in remote locations compared to more basic gravity-driven systems that can be assisted by local masons and plumbers. Additional opportunities may be provided by the introduction of smart water technologies (as discussed by Hangan *et al.* (2022)), which will have the potential to effectively monitor and manage (e.g., detecting a leak, monitoring water quality remotely). Nonetheless, their usability in the resource-bound, frequently distant environment of Nepalese CMWS depends on the factors of cost, technical capability, and connectivity, which is a major obstacle to implementation.

Community-managed water systems (CMWS) is constructed on the invisible infrastructure, which is the social fabric of a community. The social capital, trust, and common norms are concepts of great importance to the facilitation of collective action. In their research on Bangladesh, Saxena, Mowla and Chowdhury (2020) present the idea of a spiritual capital (Adhyatmik Shompatti) as a source of community welfare and sustainable tourism and note that intangible resources based on a shared culture and values are sometimes a valuable resource for collective activities.

But Nepal is undergoing sea-changing demographic changes, which are tearing this fabric. The literature is dominated by high rates of outmigration, especially of working-age men. In an explicit study, Bhawana (2020) outlines how the outmigration process impacts the livelihoods in rural communities and community forestry in Nepal by discovering that it results in loss of labour, a change in gender roles, and the undermining of the traditional institutions. This directly affects CMWS: the WUCs find it difficult to recruit members, traditional labour-sharing systems of maintenance to break down, and the financial load is concentrated on a few remaining households, frequently the aged and women. This feminisation of water management, though empowering in a way, can pose too much of a burden to women who already have too many other household and field duties.

Moreover, caste, ethnic, and gender-based social hierarchies prevent inclusive participation and continue to perpetuate inequities in water access. When some groups do not feel included in the benefits or decision-making procedures, they will not be committed to maintaining the rules of the system and will not be willing to finance it, thus resulting in its progressive deterioration.

Communities and their water system are not in a vacuum. They exist in larger socio-economic and political systems that produce opportunities as well as constraints.

External drivers play the role of moderators that may disturb the socio-economic-environmental equilibrium. Institutional factors like national water policy, Paris Agreement, technological advancement and the role of external support agencies (NGOs, donors) are the best examples of external drivers. Sustainability in a true sense occurs when such factors do not disturb the balance between three pillars: social-environmental linkages (e.g., community engagements), environmental-economic linkages (e.g., environment-friendly resource use) and economic-social linkages (e.g., fair distribution of wealth and right) (Elkington, 1998). Although it has the best intentions, donor-led initiatives may negatively affect sustainability through the cultivation of a dependence culture, the delivery of hardware without engaging in the long-term institutional development, or the use of a top-down strategy that does not appeal to local requirements (Rabah, 2003). As the analysis by Pant, Bhattarai and Basnet (2008) in the case of the Melamchi Water Supply Project demonstrates, the big and centralised infrastructure projects destroy local water management institutions and lead to new vulnerabilities.

There is market integration as well. Research on agricultural change, including that by Chamberlin and Jayne (2013) on the concept of market access and Brown *et al.* (2023) on the heterogeneous production systems of the Eastern Gangetic Plains, demonstrates the continued commercialisation of rural livelihoods. This would have two impacts on CMWS: as household incomes increase, the willingness-to-pay for water could be affected, but the competing demands on water resources in irrigation could escalate to cause conflict between domestic and productive demands.

Nevertheless, there is one serious gap. Most research studies examine one or two of these dimensions separately, such as governance in forestry or financial demand in Ethiopia. The absence of integrated and multi-metric research, which would measure the relative weight and interrelations between these factors in a specific context, namely Nepal and CMWS, is also

a problem. What is the direct effect of migration on financial viability? How far is good governance a mediating factor in the challenges of climate change? The literature purports the need to be more holistic and systemic in analysis (Mabhaudhi *et al.*, 2024).

The present study aims to address this gap by developing a unified framework in which all these multifaceted determinants are integrated. It argues that the sustainability of CMWS in Nepal can only be perceived – and eventually enhanced – when a complex set of interactions between the institutional soundness, economic sustainability, technical, and socio-cultural solidarity, and external forces are considered. In this way, it seeks to offer practical information in developing intervention strategies that are technically sound as well as institutionally robust, financially viable, and socially just.

METHODS

RESEARCH DESIGN AND DATA COLLECTION

This analysis used a mixed-methods, sequential explanatory research design to provide a detailed analysis of the sustainability determinants of CMWS in Nepal. The design consisted of sampling and analysing quantitative information, and then the qualitative information sampling followed to elaborate, expound, and contextualise the quantitative results.

The study employed a mixed-methods, sequential explanatory design conducted in three phases: 1) desktop review and case study identification, 2) quantitative data collection through household and water committee surveys, and 3) qualitative data collection through focus group discussions (FGDs) and key informant interviews (KIIs).

Four study sites (village development committee (VDCs)/ municipalities) across two districts in Nepal's middle hills were selected using stratified random sampling based on ecological zone and migration intensity. From these, 12 operational CMWS were randomly selected as case studies (Tab. 1).

Primary data collection included the following.

1. Household survey: administered to 360 households (30 per system), covering socio-demographics, service perceptions, economic factors, and community participation.
2. Water committee survey: completed by 36 water user committee (WUC) members (3 per system), assessing institutional governance, financial management, and technical capacity.
3. Qualitative data: 12 FGDs (one per community) and 15 KIIs with local officials and WUC chairs provided contextual depth to statistical patterns.

Table 1. Sampling frame

Level	Unit	Number	Selection method
District	administrative district	2	purposive (accessibility, representativeness)
Site	VDC/municipality	4	stratified (2 × 2 design)
Community	CMWS	12	random from complete lists
Household	household	360	random/systematic
WUC member	individual	36	purposive (role-based)

Explanations: VDC = village development committee, CMWS = community-managed water system, WUC = water user committee.
Source: own elaboration.

ETHICAL CONSIDERATIONS

The Institutional Review Board (IRB) of Prince of Songkla University approved this study. All participants were informed about the process of the study before data collection. The subjects were told that they had the right to withdraw at any time. All respondents received anonymity and confidentiality, and all data were stored in a secure place in password-protected servers.

DATA ANALYSIS

The SPSS v.28 and AMOS v.28 software were used to analyse quantitative data. A two-step structural equation modelling (SEM) process has been undertaken after descriptive statistics: 1) a confirmatory factor analysis (CFA) was used to justify the measurement model, and 2) the structural model was tested using hypothesised paths. Though sustainability is affected by dynamic changes yet study of static scenario by using SEM in a particular context could help to plan for future sustainability issues. Here, SEM is used to show the relation between sustainability and factors affecting it. Furthermore, qualitative information in FGDs and KIIs was transcribed and thematically analysed.

VARIABLES AND MEASUREMENT

The SEM model specified five latent constructs, each measured by multiple observed indicators (variables) from the surveys (Tab. 2).

RESULTS

DESCRIPTIVE STATISTICS AND SAMPLE PROFILE

A total of 360 households and 36 water user committee (WUC) members from 12 community-managed water systems (CMWS) participated in the study. A key demographic finding was the high prevalence of outmigration, with 68% of households reporting at least one member working abroad or in urban centres of Nepal, confirming the significance of this issue as highlighted in the literature (Bhawana, 2020) (Tab. 3).

MEASUREMENT MODEL AND RELIABILITY

The confirmatory factor analysis (CFA) confirmed the validity and reliability of the five latent constructs. All factor loadings for the observed indicators were significant ($p < 0.001$) and above the

Table 2. Latent constructs and their measurement indicators

Latent construct	Operational definition	Sample indicators (measured variables)
Financial viability	the system's capacity to generate and manage funds for O&M	regularity of tariff collection, existence of a maintenance fund, adequacy of tariff rates
Institutional governance	the effectiveness, transparency, and fairness of the WUC	frequency of meetings, transparency in decision-making, inclusivity of caste/gender
Technical capacity	the ability to maintain and repair the physical infrastructure	ease of accessing spare parts, local technical skills, frequency of major breakdowns
Community social capital	the norms of reciprocity, trust, and cooperation within the community	participation in collective maintenance, level of trust in the WUC, social cohesion
System sustainability (outcome)	the long-term functionality and performance of the CMWS	system functionality ($d\cdot y^{-1}$), water quality (user perception), user satisfaction level

Explanations: O&M = operations and maintenance, WUC, CMWS as in Tab. 1.

Source: own elaboration.

Table 3. Key descriptive characteristics of sampled households ($N = 360$)

Characteristic	Category	Frequency	Percentage
Primary livelihood	agriculture	158	43.90
	wage labour	102	28.30
	remittances	72	20.00
	business/other	28	7.80
Household member migrated	yes	245	68.10
	no	115	31.90
Satisfaction with water service	very satisfied	95	26.40
	satisfied	142	39.40
	neutral	58	16.10
	dissatisfied	45	12.50
	very dissatisfied	20	5.60

Source: own study.

recommended threshold of 0.6. Composite reliability (CR) values ranged from 0.82 to 0.89, and average variance extracted (AVE) values exceeded 0.5, demonstrating strong convergent validity and internal consistency.

Table 4. Results of hypothesis testing (direct paths)

Hypothesis	Path	Standardised estimate (β)	p -value	Supported
H1	financial viability $\rightarrow$ sustainability	0.38	<0.001	yes
H2	institutional governance $\rightarrow$ sustainability	0.42	<0.001	yes
H3	technical capacity $\rightarrow$ sustainability	0.18	<0.05	yes
H4	social capital $\rightarrow$ sustainability	0.09	0.112	no
H5	social capital $\rightarrow$ institutional governance	0.51	<0.001	yes
H6	social capital $\rightarrow$ financial viability	0.33	<0.01	yes

Explanations: β = standardised path coefficient.

Source: own study.

STRUCTURAL MODEL AND HYPOTHESIS TESTING

The structural model demonstrated a good fit with the data ($\chi^2/df = 2.15$, comparative fit index (CFI) = 0.93, Tucker–Lewis index (TLI) = 0.91, root mean square error of approximation ($RMSEA$) = 0.056). The model explained 67% of the variance (coefficient of determination ($R^2 = 0.67$) in system sustainability, indicating a strong predictive power.

The standardised path coefficients (β) testing the direct hypotheses are presented in Table 4 and visualised in Figure 1.

THE PRIMACY OF INSTITUTIONAL GOVERNANCE AND FINANCIAL VIABILITY

The results strongly confirm that institutional governance ($\beta = 0.42$) and financial viability ($\beta = 0.38$) are the most potent direct determinants of sustainability. This finding aligns perfectly with global literature on common-pool resources (Mwangi, 2017) and specifically with studies on water systems in South Asia (Ahmed Memon, 2004).

Qualitative data elucidate why governance is so critical. In high-performing systems, focus group discussions (FGDs) revealed that WUCs held regular, transparent meetings and publicly displayed their financial records. As one key informant interviews (KIIs) noted, “The system where the treasurer can explain every rupee spent is the system that never runs dry”. In

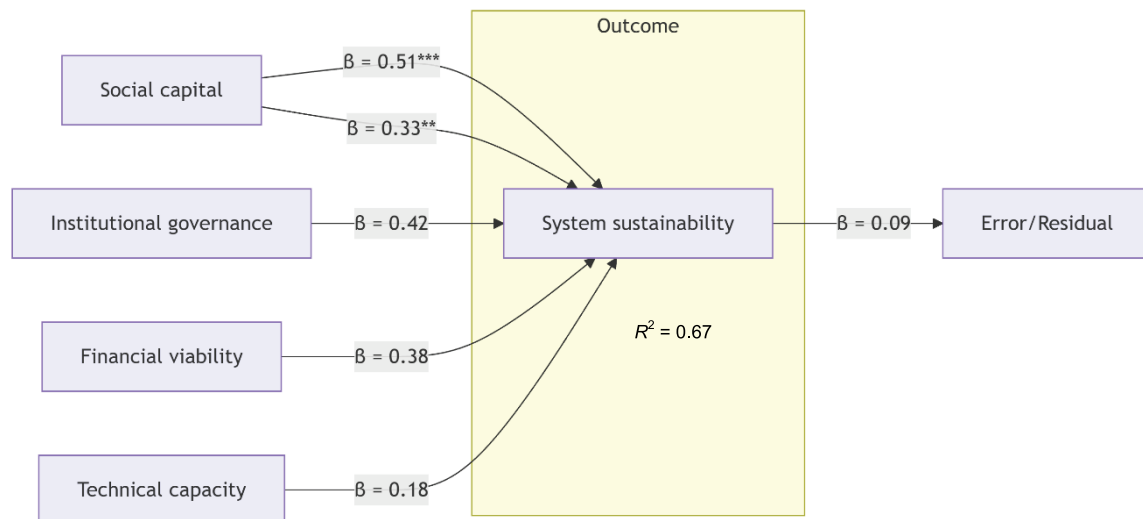


Fig. 1. Structural equation model; ** = $p < 0.01$ (significant), *** = $p < 0.001$ (highly significant), no asterisks = $p > 0.05$, R^2 = coefficient of determination, β as in Tab. 4; source: own study

contrast, systems plagued by distrust and perceptions of corruption saw sharp declines in tariff collection, directly linking weak governance to financial failure. This underscores that financial mechanisms are not just technical tools but are deeply embedded in social and institutional trust.

THE PIVOTAL INDIRECT ROLE OF SOCIAL CAPITAL

Notably, social capital had no significant direct effect on sustainability ($\beta = 0.09$, $p = 0.112$). This initially counterintuitive finding is, in fact, a crucial insight. It suggests that trust and cohesion alone are insufficient to guarantee a functioning water system. In Nepal, still water is treated as a public good, and sole responsibility is taken by the government body. The finding further indicates that social capital is being channelled through the water user committee, which interacts with the government.

However, social capital exerted very strong indirect effects by being a key driver of both good governance ($\beta = 0.51$) and financial viability ($\beta = 0.33$). The total standardised indirect effect of social capital on sustainability was 0.31 ($p < 0.01$). This means social cohesion operates as a foundational pillar – it enables the creation of effective institutions and fosters a culture of compliance and financial contribution, which in turn drive sustainability. This nuanced finding helps resolve the “paradox” mentioned by Ching (2018), showing that social capital’s value is mediated through formal structures. In the absence of strong social capital, long-term sustainability seems impossible. It might create the free rider problem as well as might create disputes among the WUCs.

THE MEDIATING ROLE OF MIGRATION

The impact of outmigration, a major theme from the descriptive results, was complex. While it eroded the structural aspect of social capital (e.g., fewer people to attend meetings), it often strengthened the cognitive aspect (e.g., remittances increased willingness-to-pay for reliable water). This is visualised in Figure 2.

This finding directly engages with the work of Bhawana (2020), suggesting that the net effect of migration is not universally negative but is instead mediated by the adaptive

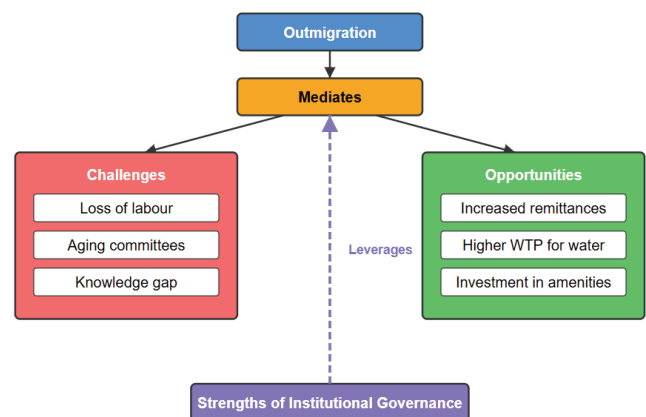


Fig. 2. The dual impact of outmigration on community-managed water systems sustainability; WTP = willingness to pay; source: own study

capacity of local institutions. Systems with strong WUCs could leverage remittances to hire paid operators, transforming a challenge into an opportunity. As an invisible force, the social capital indicator (trust and cohesion) can play an influencing role in this regard. High trust among committee members could fulfil the skill gap caused by out-migration. Furthermore, such trust could transfer the remitted fund for strengthening the WUCs in the form of a donation and a higher willingness to pay for the better water service.

TECHNICAL CAPACITY AND CLIMATE VULNERABILITY

Technical capacity had a significant but smaller direct effect ($\beta = 0.18$). It indicates that technical capacity is one of the major influencing factors of sustainability. Its direct effect seems smaller than that of the indirect effects of social capital. However, in the context of the geographical existence of the sampled area, technical capacity is a critical factor that can help to mitigate environmental and climate change-related shocks.

Qualitative data revealed that its importance is often contingent on external shocks. Systems with simpler, gravity-fed technology demonstrated greater resilience to the technical challenges posed by increasing sediment loads in water sources

(linked to changing rainfall patterns as discussed by Fatima *et al.* (2024)). Conversely, systems dependent on complex electrical pumps were more vulnerable to both climate-induced source variability and a lack of local repair capacity, supporting the argument for appropriate technology choice.

AN INTEGRATED FRAMEWORK FOR SUSTAINABILITY

The discussion of these results leads to a proposed integrated framework (Fig. 3) for understanding CMWS sustainability. The findings move beyond a list of factors to present a dynamic system where:

- social capital is a foundational driver;
- institutional governance and financial viability are the central, direct mediators;
- the impact of external drivers (migration, climate change) is filtered through the strength of these core institutions;
- technical capacity is a critical enabling factor whose importance is amplified by environmental stressors.

DISCUSSION

THE LEADERSHIP OF GOVERNANCE AND FINANCIAL SOUNDNESS: PROCESSES AND WORLD RESPONSE

The direct impact of both institutional governance ($b = 0.42$) and financial viability ($b = 0.38$) is strong, which highlights the fact that they are the key pillars of community-managed water systems (CMWS) sustainability. This observation is consistent with the world literature on common-pool resources (Mwangi, 2017) and repeats the results of studies carried out in Pakistan and Ethiopia (Ahmed Memon, 2004). The working principle here is a virtuous circle, in which good governance, which is typified by transparency, accountability, and inclusion in decision-making, will result in community trust. Such trust, in its turn, encourages the readiness to abide by the rules and pay the tariffs on a regular basis, which, directly, leads to financial sustainability.

This is supported by the qualitative data, in which systems of water user committees (WUC) with open financial records had a greater collection rate. Weak governance, fosters a sense of distrust, which leads to default on tariffs and a collapse of systems, as it has been proven to show that financial institutions have a close relationship with social and institutional settings.

Institutional governance ($\beta = 0.42$) emerged as the strongest direct predictor of sustainability, requiring transparent decision-making, inclusive participation, and accountability. Financial viability ($\beta = 0.38$) also proved significant, as rural systems frequently struggle with cost recovery and maintenance funding.

The significant direct effect of financial viability ($\beta = 0.38$) resonates with identified that community-managed systems in rural Nepal face challenges in cost recovery and maintenance funding. Oxfam's recent report noting that 71% of water supply systems in Nepal are dysfunctional, underscores the urgency of understanding these determinants (Oxfam, 2024).

THE SOCIAL CAPITAL IMPLICIT SUPREMACY

The insignificant straight line between community social capital and sustainability ($b = 0.09$, $p = 0.112$) is an important, subtle result. It implies that trust and cohesion are not enough to ensure a working water system. Nevertheless, it has strong indirect influences, through governance ($b = 0.51$) and financial viability ($b = 0.33$). The raw material out of which good institutions and fiscally responsible conduct are created is social capital. This observation can be used to address the so-called paradox of Kathmandu, described by Ching (2018) as a situation where close relationships between people prevent development in certain cases.

Our nuanced finding regarding social capital's indirect effects through governance and financial mechanisms (rather than direct effects) review of community-driven water governance across East and Central Africa identified that success requires community engagement, local knowledge integration, inclusivity, collaboration among stakeholders, conflicts resolution mechanism, as well as funding and resources mobilisation.

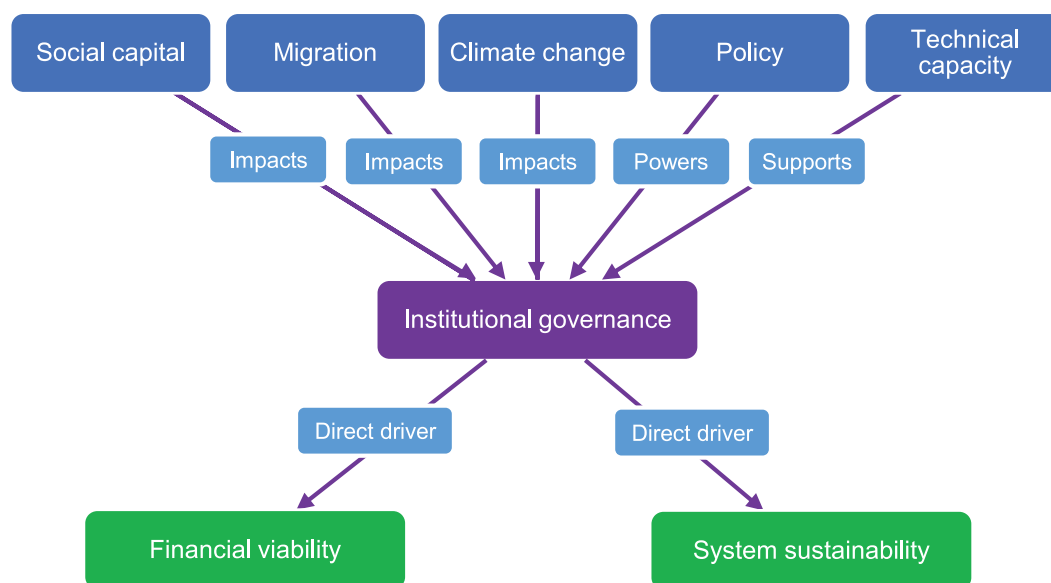


Fig. 3. An integrated framework of community-managed water systems sustainability determinants; source: own study

The research from Brazil and Tunisia examined community-managed systems in severe water-scarce areas of Brazil and Tunisia, finding that communities with strong governance could leverage external opportunities and adapt technical and organisational dimensions consistent with our mediation model (Gasmi *et al.*, 2022). A multi-country study spanning Peru, Bolivia, and Ghana, found that post-construction support and institutional strengthening were critical for sustainability, corroborating our emphasis on governance capacity building (Bakalian and Wakeman, 2009).

THE TWO-FOLD EFFECT OF OUTMIGRATION

Outmigration (68% of households) is found to be a critical demographic disruptor, which is proven by its high prevalence (Bhawana, 2020). Its effect, however, is not always negative but is rather dual and mediated by local institutional strength. On the negative, it undermines the structural component of social capital and may cause a drain of leadership. On a positive note, remittances have the potential to raise the household income and as our qualitative data reveals, the willingness-to-pay towards reliable water services. Systems that had good, adaptive WUCs could capitalise on this through the employment of more revenue to get in paid operators, hence converting a social problem into a financial problem of professionalising the system.

THE EMERGENCE OF THE NECESSITY OF COMMUNITY MANAGEMENT IN THE NEPAL CONTEXT

One of the relevant questions is why Nepal is based on a decentralised and community-managed model when a centralised system, such as water management in Singapore, is highly effective. The solution is presented in contextual differences. Singapore is a high-density, urbanised city-state that possesses a powerful centralised government and has good financial strength. By comparison, the rugged terrain and the location of settlements in Nepal are so scattered that it is technically precluded and astronomically costly to build and maintain a centralised grid. Moreover, the fact is that the state has little ability to control such an enormous and complicated network in remote places, and hence the model of community ownership does not necessarily happen because it is preferred, but rather a necessity. Hence, the research inquiry and policy should not be to substitute CMWS but comprehend them and reinforce them as the most feasible approach towards attaining rural water security in Nepal.

CONCLUSIONS

The research paper manages to draw the conclusion that the sustainability of community-managed water systems (CMWS) in Nepal is largely defined by the capacity of the local institutions and their financial well-being, as opposed to the technical and social factors alone. Social cohesion is vitally important as it offers a criterion basis, but it is achieved indirectly through the efficient administration of governance and promotion of fiscally responsible behaviour. The widespread issue of outmigration has a dualistic effect, which on the one hand undermines the aspect of community inclusion and on the other hand has the possibility of increasing financial resources through remittances. Finally, the

resilience of a system to manoeuvre through these complexities and also to withstand external shocks such as climate change is mediated by the strength of its water user committee. The main recommendation, which is based on the findings of the critical role of institutional robustness and financial viability as the key direct determinants of sustainability, is to reform intervention strategies towards integrated capacity-building programs, which will not only enhance the governance capacity of water user committees but also financial management systems. This is to include training in open accounting, strategic planning, as well as inclusive leadership, and formalising and computerising the management of tariff collection and maintenance funds to enhance efficiency and trust building. Through their ability to establish a systematic relationship between good governance and financial resilience, such programs would enable communities to cope better with external stressors such as migration and climate change and, as a result, enhance the long-term operational sustainability of their water systems. For water user committees, we recommend institutionalising financial transparency through public quarterly reports, establishing dedicated maintenance funds with 20% tariff reserves, and leveraging remittances from migrant members through annual contribution options. For municipal governments, we recommend establishing tiered technical support units, creating incentive mechanisms linking co-funding to governance benchmarks, and facilitating inter-community learning networks. These evidence-based recommendations operationalise Nepal's National Water Supply and Sanitation Policy while addressing key determinants our model identified as most influential.

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CONFLICT OF INTERESTS

All authors declare that they have no conflict of interests.

INSTITUTIONAL REVIEW BOARD STATEMENT

This paper has been conducted in accordance with the ethical standards. The Institutional Review Board (IRB) of Prince of Songkla University gave ethical approval.

DATA AVAILABILITY STATEMENT

The data produced and processed in the present research cannot be accessed publicly because they were protected by confidentiality agreements between the respective communities; however, upon request, we could make them available.

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